

Suryadipta Ghosh

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Skills

- **Languages:** Python, JavaScript, TypeScript, C++, SQL, HTML, CSS
- **Web:** Node.js, Express.js, Next.js, React.js, MongoDB, PostgreSQL, GraphQL, REST APIs, JWT, WebSockets, Docker, Redis
- **ML/Data:** NumPy, Pandas, scikit-learn, XGBoost, SHAP, Matplotlib
- **Tools:** Git, GitHub, Postman, Vercel, GitHub Actions, Google Colab, Jupyter

Experience

Research Intern *CSIR-CSIO, Chandigarh* Jan 2026 – May 2026

- Benchmarked **5 ML regression models** (SVR, KNN, Random Forest, XGBoost, BPNN) for power estimation of VLSI circuits on ISCAS'89, applying **LOOCV** to surface overfitting missed by conventional splits; **XGBoost** achieved best generalization with LOOCV R^2 of **0.905**.
- Applied **SHAP TreeExplainer** and permutation importance to tree-based models; identified total gate count, inverter count, and D-flip-flop count as dominant predictors, with gate-output interaction as the strongest pairwise effect.
- **Tools:** Python, scikit-learn, XGBoost, SHAP, NumPy, pandas, Matplotlib

Projects

Svaralike – AI-Powered Semantic Music Discovery Engine [\[GitHub\]](#) [🔗](#) (*Ongoing*)

- Engineered a semantic search backend converting mood queries into embeddings via **Hugging Face**, retrieved with **pgvector**, re-ranked using **Spotify** audio features, and explained via **Gemini API**.
- Built a **Node.js/Express** REST API with **Redis**-cached, **BullMQ**-queued audio-feature processing, containerized with **Docker**.
- **Tools:** Node.js, pgvector, BullMQ, Hugging Face, Gemini API, Docker, Redis

DevLog – Developer Activity Dashboard [\[GitHub\]](#) [🔗](#)

- Built a full-stack analytics platform on the **GitHub REST API** with dual **REST + GraphQL** layers, **Mongoose** models, **JWT** role-based auth, and **GraphQL Subscriptions** over WebSockets for real-time star notifications.
- Added a **Transform Stream** CSV export pipeline and **Cluster module** entry point for multi-core utilization.
- **Tools:** Node.js, MongoDB, Apollo Server, JWT, WebSockets, GraphQL

QueryForge – PostgreSQL Query Runner [\[GitHub\]](#) [🔗](#)

- Built a **connector-based datasource interface** for **PostgreSQL** with a **NestJS** API exposing query execution endpoints, and a **React** dashboard for running queries and viewing results in a live table.
- **Tools:** TypeScript, NestJS, React, PostgreSQL

Education

Punjab Engineering College *B.Tech, Electronics Engineering (VLSI Design and Technology)* Aug 2023 – May 2027

- CGPA: 6.78

Coding Profiles

- **LeetCode** – 100+ DSA problems solved (arrays, strings, trees, graphs, DP) leetcode.com/sxryadipta [🔗](#)
- **GeeksforGeeks** – 50+ DSA problems solved geeksforgeeks.org/sxryadipta [🔗](#)

Publications & Open Source Contributions

- **IEEE CHANDICON 2026** – Conference paper, "A Comparative Study of Interpretable Machine Learning Models for Power Estimation of VLSI Circuits", accepted for oral presentation at IEEE CHANDICON 2026.
- **ToolJet (Open Source)** – Opened PRs adding a new **Plivo "Make Call" operation** to the marketplace plugin system (TypeScript, Node.js) and fixing a missing **Jest configuration** gap in the plugin test suite, following the project's contribution and review workflow.
- **StatsBomb Open Data** – Contributed Python-based data validation scripts and a **GitHub Actions CI/CD workflow** to automate verification of football event datasets; identified and reported data quality issues through pull requests and issue tracking.